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Trade-offs?: Long-run Effects of Trade Liberalisation Reforms on Local Employment, Marriage and Fertility in Urban India

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Abstract

This project explores the impact of the 1991 trade liberalisation reforms on regional outcomes in Urban India. Through the drastic reduction in tariffs across industries, I consider how the long-run effects on regions differed with a shift-share instrumental variable approach. As reforms serve as an exogenous labour market shock, I offer insights into how changes in employment affect family structure decisions. While I do not find a notable change in men's employment on average, an increase in one standard deviation of a region's degree of exposure to trade liberalisation corresponds to a 14% decrease in female employment, and a 4% increase in the marriage rate for 20 - 24 year-old women. I consistently find this negative relationship between mechanisms affecting women's employment and early marriage rates, supporting the idea of an 'independence effect' for women in Urban India. Estimates by gender-specific exposure point to considerable heterogeneity within regions on how workers in different sectors were impacted, as does considering the contribution of shocks to individual industries.

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1 Introduction

Reductions in countries' barriers to trade are often seen as a central part of modern development efforts, having historically influenced the economic trajectory of many economies, notably the East Asian tigers. A large wave of developing countries underwent extensive trade liberalisation reforms in the 1980s and 1990s. India's reforms in 1991 featured a sharp reduction in tariffs of 52% on average across all products (unweighted), and broader shifts away from a highly protectionist model of trade, driven externally by the International Monetary Fund (IMF), as a condition for their support with India's balance of payments crisis.

A key question is how the effects of reforms varied by different subgroups of the population, and across regions, to consider the consequences of globalisation on widening economic disparities. This project analyses how the degree of exposure to trade liberalisation reforms affects long-run changes in employment, marriage and fertility in urban India at a regional level, comparing outcomes between 1991 and 2001. In addition to offering insights into the direct effects of increased trade, I contribute to exploring causal relationships between labour market outcomes and family structure dynamics, since India's trade liberalisation reforms can be exploited as a quasi-natural shift to the Indian labour market, navigating standard endogeneity concerns.

Regions' varying exposure arises from distinct industry sectoral compositions of employment, with sectors experiencing different degrees of reduction in the average tariff of related goods. This setting motivates a shift-share instrumental variable design, to consider the differential impacts of common national shocks based on a region's pre-determined exposure. With two pathways to satisfying the exclusion restriction of a shift-share instrument, I assume exogeneity of the shocks; that between 1991 and 1997, tariff changes were not systematically higher or lower in industries concentrated in regions which experienced shocks to labour supply and factors influencing marriage and fertility. This is supported by the rapid, external implementation of liberalisation reforms weakening standard correlations between industry characteristics and tariff changes, also examined in prior literature.

To gauge deeper insights into relative effects between men and women in addition to overall regional exposure, I construct gender-specific shift-share instruments, considering the effect of a gender's exposure to tariff reductions in a region. Since I provide reduced form results as opposed to considering a general equilibrium model, my results provide relative comparisons of regions in India rather than trying to reach an overall conclusion of whether trade reforms were beneficial to the economy and people at an aggregate level.

I find that in regions where workers are more exposed to liberalisation reforms, overall female employment decreased, while marriage rates for 20 - 24 year-old women modestly increased, statistically significant at the 5% level. The effects of female-specific exposure, controlling for male-specific exposure and the share of women working in 'traded' industries, moves in a contradictory direction which suggests at heterogeneous effects for women in traded compared to

non-traded industries. Effects on men’s employment on average are small and statistically insignificant, however there are opposing, significant effects from tariff reductions to the industrial manufacturing industry and wholesale, retail and trade.

The rest of the paper is organised as follows: Section 2 describes existing related literature and the context of Indian trade liberalisation reforms. Section 3 considers the theoretical intuition behind ex-ante hypotheses, linking employment and marriage decisions. Section 4 describes the regional and tariff data used. Section 5 outlines the construction of the shift-share instrumental variable regressions and necessary exclusion restriction. Section 6 presents the findings, before considering mechanisms and robustness checks in 7, and concluding in 8.

2 Relevant background

2.1 Existing literature

I draw on contextual information to motivate a shift-share instrumental variable strategy from Petia Topalova’s work, which considers the impacts of Indian trade liberalisation reforms on regional poverty and firm productivity from 1991 - 1997 (Topalova 2004, Topalova, 2010). In considering the effect of gender-specific components of labour market shocks, my approach to decomposition is similar to Autor et al. (2019) who estimate how shifts in the relative earnings and employment of young men compared to young women, arising from increased trade pressure on manufacturing in the US, impacted marriage, fertility, and children’s living circumstances.

While focusing on rural poverty which was found to have increased with exposure to trade liberalisation reforms, Topalova found poverty rates declined faster in urban areas with a greater tariff exposure. Topalova and Khandelwal also document at the firm-industry level that on average, a 10% reduction in industry output tariffs (for the final goods produced) raised firm total factor productivity by 0.53%, as a result of both pro-competitive forces in final goods driving greater efficiency and access to better production inputs due to lower input tariffs. Effects were strongest in import-competing industries.

There is a relatively growing literature considering the impact of trade liberalisation reforms from a gendered perspective in the context of developing countries, which often have social norms unique in their degree of restrictiveness and discrimination on the basis of gender. Gaddis and Pieters (2012) find tariff reductions in Brazil to have been associated with short-run female labour force participation increases, attributed to greater labour market insecurity and male unemployment as mechanisms, through which women join the labour force to smooth household supply. Juhn et al. (2013) similarly find trade liberalisation to increase women’s average relative earnings and employment in Mexico, through blue-collar jobs in particular, where the introduction of new technologies reduces the value of ‘brawn’ and so diminishes preferences in hiring men. A broader literature considers the impacts of other exogenous labour market shocks which had differential exposure to genders, on relative outcomes for boys’ and girls’ survival rates and education (Qian 2008; Heath and Mushfiq Mobarak 2015).

To the best of my knowledge, a job market paper by Shruti Sengupta (2019) considers a research question most similar to my project work, examining the effects of trade liberalisation reforms in India on marriage and fertility. My approach differs and adds in contribution by analysing the effects of gender-specific trade shocks and considering the impact of individual industry groups, which offers greater insights into the heterogeneous relative effects for different groups of workers. I formally follow a shift-share instrumental variable strategy, considering best practices when assuming exogeneity of the instrument through the shocks to shares, formalised by Borusyak et al. (2024) and Adão et al. (2019).

2.2 Context of trade liberalisation reforms

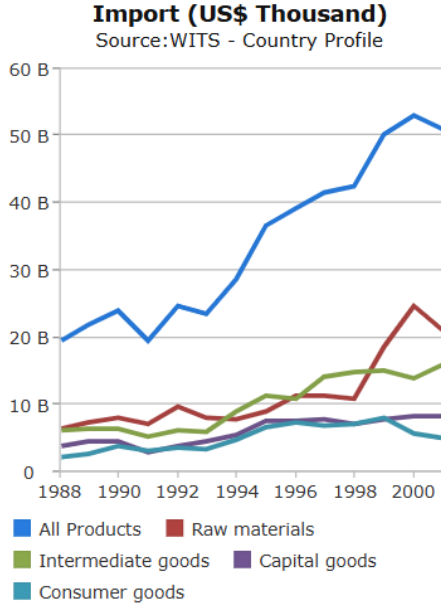
India pursued a strategy of import-substitution-led economic development titled ‘Make in India’, and self-reliance for forty years following post-independence, before the large-scale reduction of tariffs, industrial de-licensing and devaluing of the rupee (Kirmani and Wong 1997). India’s broad, industry-general trade liberalisation reforms beginning in mid-1991 were largely driven by conditional support from the IMF in tackling the balance-of-payments crisis. The average tariff rate across products dropped substantially, from a trade weighted average tariff of 56.4% in 1990 to 20.3% in 1997¹. Changes to non-tariff barriers included the proportion of products subject to quantitative restrictions falling from 87% to 45% between 1987 - 1995, and an 18.7% depreciation in the exchange rate.

This rapid and externally imposed reform lends itself to empirical study, since it plausibly navigates endogeneity concerns of trade policies by industry. Topalova provides evidence of tariffs being uncorrelated with industry characteristics until 1997 (1992 – 1997 indicates an episode of India’s 5 Year Trade plan)². Following this, tariffs in India rose to a limited degree across most industries, with a weighted average tariff of 25.5% in 2001. Between 1997 – 2001, policymakers were more selective in setting product tariffs, and Topalova finds a statistically significant negative correlation between industry productivity and tariffs after 1997, where less efficient industries were protected.

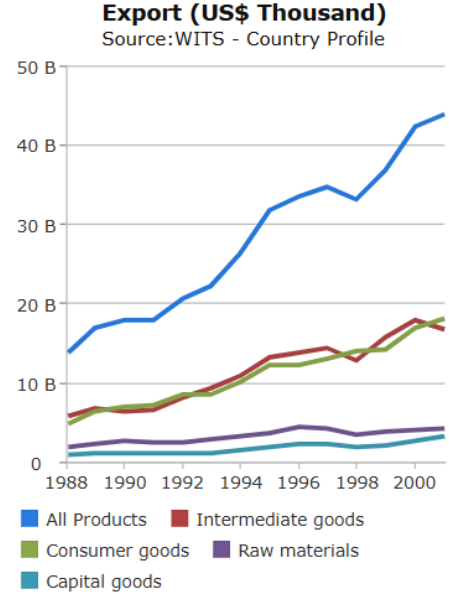
The graphs in Figure 1 show the changes in the trends of value of imports and exports from 1988 to 2001, generally following similar upward patterns by broad main product groups, although there are differences in the trends for raw materials and would likely have greater heterogeneity by individual products. A reciprocal relationship between increased imports and exports is supported by the idea of a country’s import restrictions often acting as a tax on their exports, where protection of domestic industries can frustrate the goal of increasing exports (Tokarick 2007). My analysis leverages changes in import tariffs as shocks to industries; if there were significant tariff changes applied to Indian exports, analysis by changes in average export

¹The simple average tariff across all products was 80.9% in 1990, 28.9% in 1997, and 31.8% in 2001. Weighting accounts for the volume of the good traded.

²I note here that to satisfy instrument exogeneity, the assumption required does not have to be quite as strong as tariff changes being completely random across industries (this would be a sufficient, stronger condition) - discussed in 5



(a) Imports



(b) Exports

Figure 1: Value of Imports and Exports (reported by India)

tariffs would reflect the labour market effects of increased foreign demand for Indian products that a region experiences. Given India's largest export partner shares in 1991 were for the United States, the Soviet Union, Japan, Germany and the United Kingdom which already had low applied tariff rates on Indian goods, the rise in exports by industry³ does not appear to have risen from a reduction in export tariffs (from World Integrated Trade Solutions).

Rather, firm productivity increasing through production inputs becoming cheaper and more varied, as found in previous literature, would have increased global competitiveness and could be linked to greater exports. De Loecker et al. (2016) relatedly find firms experienced a 31% reduction in marginal costs on average, and while not passing this through entirely to the consumer and instead increasing their markups, the price of Indian goods on average also reduced by 18%. Thus, my analysis of regional outcomes by degree of trade protectionism using import tariffs as shocks, may also reflect in part the knock-on effects these industries faced from similar increases in export demand.

Alongside the reduction in tariffs, India simultaneously implemented a number of complementary economic policies to Non-Tariff Barriers to facilitate greater trade, including abolishing a previous 40% threshold on foreign equity investment and industrial de-licensing (Virmani 2001). If any industry-specific policy changes are complementary with the tariff change for that industry, my analysis would offer insight into the effects of these broader changes to trade openness, in addition to tariff reductions.⁴

³The devaluation of the rupee, while an export-promotion strategy, should not drive heterogeneous effects by industry groups.

⁴I would not expect complementary policies to act as a confounder to my shift-share strategy, if externally implemented

3 Theoretical intuition

From the findings of Topalova and Khandelwal (2011) and De Loecker et al. (2016), we might expect a reduction in marginal costs and increase in productivity to have resulted in increases in long-run labour demand, as firms sought to scale and increase output. Likewise, the Stolper-Samuelson theorem, within the framework of Heckscher-Ohlin trade theory hypothesises that trade liberalisation increases returns to the factor more intensively used in production and which a country is comparatively abundant in (i.e. labour for India, as in most developing countries). However, if the new technologies introduced via imports are labour-saving, employment might decrease as it is substituted for capital. The returns to different factors of production would likely differ in sectors depending on production processes, and the degree to which these changed.

Since trade theory is not conclusive on the effects of employment or sectoral reallocations, in the following intuition I consider trade liberalisation to be an ambiguous shock to household members' employment. Female labour force participation has been markedly low and stagnant in India, meaning all industries are male-dominated, however different industries attract varying proportions of working women (Figure 9 and Table 0.1). Most notably, the 'Other Services' sector, which while referred to as non-traded throughout would still likely experience effects in regional general equilibrium, accounts for 42 - 45% of female employment. Hence, gender-specific shocks to employment can be motivated, for both existing workers and for potential entrants joining the labour force. I reflect on how the dynamics of household bargaining and decision making may be altered, abstracting from the specific dynamics trade might potentially have. Since I consider outcomes with a gap of ten years in my empirical analysis, the intuition captures a static snapshot of a more dynamic process - where households respond to a new equilibrium environment.

3.1 Collective household decisions

Within a household, I predict a converse relationship between changes in partners' employment; if there are negative effects on men's employment, this potentially may lead to positive effects for women's employment through the 'added worker' effect. If trade liberalisation were to harm men's employment, this would act as an income shock to a household, and potentially spur women to join the labour force to smooth household consumption, empirically documented in business cycle downturns in developing countries (Verick 2014). Conversely, if her partner were to see an improvement in employment prospects, this could reduce the probability of her working through a household income effect. Hence, a women's likelihood of participating in the labour force will be a function increasing in her employment offer, and decreasing with respect to her husband's offer and the social cost of her working, which would likely be higher in particular industrial occupations such as factory-based manufacturing (Jayachandran 2020).

alongside tariffs - the composite shock would need to satisfy the appropriate shift-share exclusion restriction. In addition, formal licensing requirements were relinquished across all industries apart from for safety reasons - much of the policies appear to be industry-general (Kirmani and Wong 1997).

The social cost to working women in the family is likely endogenous; as female labour force participation grows, the disutility from social norms of women working falls - for instance, a son with a working mother growing up is less likely to have the same level of distaste for his wife working as he otherwise might have had, hence why we might expect long-run trends to amplify and exacerbate initial impacts.

3.2 Deciding to marry

Becker (1973) formalises the intuition behind relative gendered differences in economic stature affecting marriage rates, in his seminal Theory of Marriage.

Men and women, both when single and married, allocate their time between market and non-market sectors, seeking to maximise total output. Becker hypothesises that a rise in women's wage rate (w_f) relative to men's wage rate (w_m), decreases the gains from marriage if at baseline $w_f < w_m$. A key assumption this relies upon is that women's marginal productivity of labour in the non-market sector is greater than that of men (and lower in the market sector), thus resulting in each spouse's specialisation towards their comparative advantage when married. While this assumption is controversial, we can empirically rely on the result in the Indian setting that on average, a women works fewer hours in the market sector when married, compared to when single, conditional on observable characteristics (Das and Desai 2003). Thus, the smaller the gap between w_f and w_m , the lower the gains are from substituting a man's time in the market for women's time, and hence marriage is less attractive. While labour returns are quantified here as wages, and I observe changes to employment but not earnings, a rise in female employment can proxy for improved economic opportunities for women or worsened opportunities for men (by revealed preference, since more women are choosing to work and are hired).

Strong contextual factors around marriage in India make it important to deviate from the benchmark model of marriage in Becker's Western context and factor additional considerations. Becker includes a brief discussion of when the likelihood of women participating in the labour force regardless of marital status is low. In this setting, gains to marriage are larger since outside options to domestic non-market work are reduced. However, if opportunities to earn start to arise, the predictions made above begin to materialise. It is also more likely, that for a significant proportion of couples, rather than agents independently deciding whether or not to marry, this would be joint household decisions by the bride and groom's family. Similar logic extends to these decisions, but delays in marriage come at an added social cost to women and their families, often through criticism, ridicule, and difficulties in marrying later - a higher relative w_f will not necessarily guarantee a decline in marriage rates, factoring in offsetting social costs.

An additional consideration potentially contradicting the stylised prediction from Becker's model is the prevalence of dowry payments⁵, which has been suggested in the literature as

⁵Payments made (often through a transfer of assets) as part of an informal marriage 'contract', typically from the bride's family to the groom's. Although the practice was deemed illegal in India in 1961, it unfortunately remains prevalent,

a mechanism which might potentially offset the independence effect above, where a rise in a women’s income could result in increased abilities of families to make sufficient dowry payments and thus increase the rate of marriage. Ex-ante I believe that particularly in the long-run, the independence effect dominating the income-dowry effect is more likely. If women persistently have relatively higher economic opportunities in regions more exposed to trade liberalisation, dowry demands and concessions may also plausibly decline, through the elevated economic stature of women. Given the extortionately high sums demanded by families practising dowry payments, it is plausible that majority of the wealth transfer would also be facilitated by the household’s wealth, as opposed to the bride’s income.

Put together, I make the following proposition (2): *if exposure to trade liberalisation raises male employment opportunities more than that of females, earlier marriage for women becomes optimal. If trade raises female employment opportunities more than male, delayed marriage may be more optimal, depending on the magnitude of social cost.*

4 Data

4.1 Regional data

The data used for analysis is collated from district-level census datasets and tariff data to form a balanced panel. The decennial census provides data on main workers’ employment by industrial category, sex and urban/rural regions at the district level in 1991 and 2001, in addition to total population figures. Each district belongs to a state or union territory, of which there are 31 in 1991, with district and state sizes varying considerably by area and population (see Summary Statistics in Appendix). Employment is disaggregated into ten industry categories: cultivators; agricultural labourers; manufacturing in the household industry⁶; livestock, forestry and hunting; mining and quarrying; manufacturing, processing and repairs; construction; trade and commerce; transport, storage and communications; other services. Marriage and fertility data, by age groups of women, is also obtained from the census. The share of women married and number of women who gave birth in the past year are reported for currently married women; given the very low rates of divorce and childbirth outside of marriage in India during this period, this is representative for the total population of women.

District boundaries changed substantially in India from 1991 to 2001, owing to districts partitioning (there were a total of 466 districts in 1991, and 593 in 2001) and tehsils (unit of administrative division) being transferred between districts. To construct my panel, I identify

alongside dowry-related violence, with an estimation of 25,000 related deaths and maiming of women aged 15–34 in India every year (Banerjee 2014). While official estimation is largely missing, prevalence of dowry demands is common across both rural and urban areas; a survey of approximately 10,000 women found dowry demands of 26% in non-slum urban areas in 1998-99.

⁶Defined by census as: industry conducted by head of the household and members of the household or within village. Not run on the level of factory

387 composite regions as close to the district-level as possible with unchanged boundaries from 1991 to 2001, following Census Appendix guides on changes and Kumar and Somanathan (2009). For example, if district A transferred a tehsil to district B in the time frame, in my analysis they are aggregated to one region⁷. Where a district partitions, or changes name, I keep to using the 1991 name of the original district. Similarly, three new states forming from separations of regions in 2000 are assigned to the 1991 parent states. Henceforth, I refer to the level of analysis as ‘region.’ The union territory of Jammu and Kashmir (approximately 20 districts) is excluded from my analysis, given political conflict prevented data collection in 1991.

Rural parts of regions are omitted from my analysis, given my instrument was weak with a first-stage F-statistic of 8.76, largely driven by the relatively larger disparities in instrumented and treated tariffs for agricultural goods⁸. Contrary to other trade settings where rural production might be thought of as the most traded products, India canalised a large proportion of commonly grown agricultural produce until 1999, restricting the trade of wheat, cereals, rice and oilseeds (Chand and Bajar 2012)⁹. 70% of workers in rural regions in 1991 were involved in the cultivation of cereals and oilseed (Topalova 2010)¹⁰, hence if rural regions were included despite instrument concerns, majority of workers would have been classified as working in a non-traded sector and not as part of the shift-share treatment.

Outcome variables: Regional employment, as a proportion, is measured as the total main worker population over the total population subtracted by the number of non-workers classified as dependents and retirees¹¹. Due to some potential ambiguity in the definition of ‘dependent’, I include employment results not excluding dependents and retirees from the total regional population in Appendix A2, to which my results are robust. The total main worker population is the sum of main workers across all industry groups. 1991 employment data is used for pre-determined industry shares prior to the treatment period, given the census was completed by April 1991 and trade liberalisation reforms began in June 1991, so would not have been affected by the shocks. Shares as close to the treatment start date as possible also maximise the relevance of the instrument. A key limitation is not having access to average income data at the district-level, which would offer a more detailed insight in being able to capture the intensive margin of employment and labour market opportunities as opposed to solely the extensive margin. I note that considering the proportion of main workers of the region’s total population as my outcome of interest does not allow me to comment on broader trends in the number of unemployed job-seekers in a region. Since ‘main workers’ are classified as those who are reported to be working for at least six months of the year, this likely misses a large share of short-duration and casual

⁷The same applies for multiple tehsil transfers between multiple districts

⁸Mean joint share of workers in agriculture and as cultivators in rural regions: 0.778 (s.d. 0.145). There is some reduction in the share of workers in cultivation and agriculture between 1991 and 2001, however this still remains high in rural regions in 2001

⁹i.e. exported only through agencies designated by the central government.

¹⁰80% of workers were in agriculture, of which 87% were cultivating cereals and oilseeds

¹¹Dependents are classified in the census as ‘including all dependents such as infants/children not attending school or permanently disabled from work due to illness/old age. May include able-bodied people who cannot be categorised in any other category of non-workers but are dependent on others.’

informal workers, leading my analysis to exclusively look at how reforms affected formal market work. As an outcome, this is not necessarily a limitation since trends in wage work in the formal sector inform of patterns in higher quality and more secure employment, however if informal workers are more concentrated in particular sectors, I may not fully capture a region’s exposure to tariffs which could have affected regional general equilibrium outcomes.

I focus on the share of women married between ages 15 - 19, and 20 - 24, and the share of women aged 20 - 24 years who gave birth in the past year, as gendered socioeconomic outcomes (refer to Appendix for marriage and fertility results across different age bands and all ages). These particular ages are chosen for marriage, since 87% of women in urban regions are married by age 24, hence to track variability in changing attitudes, earlier marriage rates would likely be more responsive.

The fertility measure is defined as the share of women in the given age range who gave birth in the past year, where births in India in both 1991 and 2001 were most likely to be from women aged 20 - 24. The mean share of 15 - 19 year old women giving birth in my dataset is 0.023, hence I do not think changes would reflect attitude shifts and responses as broadly as the 20 - 24 age range.

4.2 Tariff data

Data on import tariffs rates is obtained from World Integrated Trade Solutions (WITS). The WITS is composed of disseminated trade and tariff data from UNCTAD Trade Analysis and Information System (TRAINS), classifying and averaging approximately 5000 6-digit product lines into 27 aggregated groups, with the 1988/92 Harmonised System (HS) classification codes available for all products encompassed by each industry group - for example, ‘Food products’ product group contains HS product codes 16 - 24 (by the starting 2-digits). To assign products to the census industry categories, I referred to a concordance table of 1988/92 Harmonised System (HS) classification codes of products to International Standard Industrial Codes (ISIC) for employment industries as a general guide, given concordance directly to National Industry Codes (NIC) for India was not readily accessible.

This did entail some arbitrary categorisation, where a product group’s codes would overlap across two census industries - for example agricultural labourers and cultivators are assigned as facing the same tariffs, as are manufacturers and those employed in manufacturing industries in the household industry.¹²

For my analysis, I use the weighted Effectively Applied AHS (Ad-valorem, Harmonised System) tariffs measured in % of the product’s value. The AHS is defined as the lowest available tariff – if a preferential tariff exists, it will be used as the effectively applied tariff. Effectively

¹²The difference between cultivators and agricultural labourers is defined as: cultivators work on their own land, whilst agricultural labourers work on others’ land for wages. The NIC codes under the household industry breakdown also included manufacturing of chemical products, metals, electrical components, general purpose machinery, in addition to other product lines shared by the manufacturing industry.

Applied Tariff and Most-favoured Nation (MFN) tariff values have a correlation of 1 in both 1991 and 1997, and a correlation of 0.998 in 2001 - which is representative across industry groups, suggesting my analysis is not sensitive to the choice between these tariffs most commonly referred to in analysis.¹³ Weighted tariffs account for the volume of product being traded, where weights are assigned to different products based on what a country typically imports, reflecting that some tariffs might be relatively high but not significantly impact trade if being applied to low-volume or negligible products. Products frequently imported get assigned bigger weights.

5 Empirical approach

Equation (1) presents the OLS fixed-effects baseline specification of interest, with outcomes regressed on a region's degree of exposure to tariffs, as a metric of trade protectionism. L denotes the number of main workers in region d working in industry j at time t . $\left(\sum_j \frac{L_{djt}}{L_{dt}} \cdot \text{Tariff}_{jt}\right)$ is mechanically a measure of the region's degree of trade protectionism by tariffs; the industry-employment share of workers in the region interacted with the common national tariff for that industry, summed across all industries. With two time periods, my two-way fixed effects regression is equivalent to a long-differenced specification with a gap of ten years; time fixed effects account for any broader common macroeconomic shocks, and region fixed effects control for time-invariant heterogeneity in regions.

$$Y_{dt} = \alpha + \beta \left(\sum_j \frac{L_{djt}}{L_{dt}} \cdot \text{Tariff}_{jt} \right) + \gamma_d + \lambda_t + \epsilon_{dt} \quad (1)$$

There are two primary endogeneity concerns in this specification. First, we might expect industry-employment shares to react to industry tariff shifts and change accordingly in the next period. This would be problematic and bias our estimates if this occurred in a way correlated to the error term (Borusyak et al. 2024). For example in the case of employment as an outcome, suppose the reallocation of employment is towards industries with relatively smaller reductions in tariffs. Regional labour market regulation, influencing flexibility, will determine to what extent this reallocation happens, and is also a factor influencing employment rates (Besley and Burgess 2004). Hence, in the next time period, regions with greater labour market flexibility would on average have a higher treatment value (workers are relatively more protected, having moved from industries with presently lower tariffs to higher tariffs). This motivates the use of pre-determined industry employment shares, lagged to the beginning of the treatment period, to instrument for employment shares in all time periods. 1991 employment shares satisfy being pre-treatment measures, given the completion of the census before reforms began. Shares taken

¹³Countries promise to impose MFN tariffs on imports from other members of the WTO (India became a member of the WTO in 1995), unless the country is part of a preferential trade agreement. In essence, the MFN rates are the highest tariffs WTO members can charge each other.

closer to the start-of-treatment date lend themselves to constructing instruments with greater relevance.

Second, as outcomes are considered in 1991 and 2001, this extends beyond the period of ‘as-good-as-randomly assigned’ tariffs to industries. If tariffs are systematically different based on industry characteristics, they may also be correlated with regional confounders, where industries are clustered in particular regions. I therefore also instrument 2001 tariff values with 1997 values, given they move in the same direction across all industry groups with the exception of wholesale, retail and trade¹⁴, leading to a strong overall relationship; the correlation between the changes in average tariffs from 1991 to 2001, and 1991 to 1997, is 0.94.

5.1 Shift-share instrument

The standard assumptions behind instrumental variable approaches apply. Relevance of the SSIV instrument requires that in addition to 1997 and 2001 tariff movement correlation, employment-industry shares persist to a sufficient degree from 1991 to 2001, both of which are met through a sufficiently strong first-stage (Section 6). Exogeneity requires that no other local factors influencing employment, or marriage and fertility decisions, are correlated with national tariffs from 1991 to 1997 weighted by a region’s baseline employment shares. I discuss monotonicity below.

Pre-determined industry-employment shares and tariff instrumentation result in the following reduced-form SSIV analysis:

$$Y_{dt} = \alpha + \beta \left(\sum_j \frac{L_{dj1991}}{L_{d1991}} \cdot \text{Tariff}_{j1991,1997} \right) + \gamma_d + \lambda_t + \kappa \left(\sum_{j \neq OS} \frac{L_{dj1991}}{L_{d1991}} \right) \delta + \epsilon_{dt} \quad (2)$$

‘OS’ denotes the ‘Other Services’ industry, encompassing financial intermediation, business activities, education and health services, and is the only industry assigned a tariff of 0. This endogenously limits the size of the instrument across regions, whereby regions with a higher Other Services share could plausibly have a lower instrument value on average, but higher employment rates from regional infrastructure and labour market characteristics complementary to the industry, biasing our coefficient estimate on exposure to trade protectionism (the Other Services industry has the highest share of graduate workers, with over 25% of the workforce having at least a graduate degree - see Figure 8 in appendix). Regional fixed effects absorb the sum of baseline shares, however this insufficiently captures the full effect. Regions with a higher ‘OS’ share, and so lower instrument share subject to the change in tariffs, are likely to have a higher change in the error term over time compared to other regions, perhaps arising from increased productivity and human capital in financial services, education and IT industries.

Hence, κ denotes a dummy indicating the second time period (2001). This is interacted with the baseline sum of industry-employment shares to capture the differential impact the sum of

¹⁴The average tariff for wholesale, retail and trade decreases between 1991 - 1997, and rises from 1997 - 2001, in line with the pattern for all other industries. However, starting from a uniquely low baseline tariff in 1991, the later tariff rise means average tariffs were higher in 2001 than 1991.

shares has in the second period; equivalent to controlling for a linear time trend in the sum of baseline shares. The equivalence to a long-differenced specification is still maintained; this control would be the sum of baseline shares alone. Borusyak et al (2024) show controlling for the sum of shares is often superior to renormalising shares to sum to 1 (the denominator for shares would be the sum of workers excluding ‘Other Services’ workers), as otherwise done in the approach pioneered by Topalova, for greater statistical power.¹⁵ Controlling for the sum of baseline shares means the SSIV now represents a weighted average of shifts for a region, as opposed to a weighted sum of shifts.

Standard errors are clustered at the regional level, however I discuss the potential over-rejection this might introduce in lieu of computing exposure-robust standard errors. While my outcomes range from 0 - 1, I do not estimate my regressions with a probit or logit model, since shift-exogeneity does not imply exogeneity of non-linear transformations of the SSIV (Borusyak et al. 2024). I check to confirm my predicted values stay between 0 and 1.

5.2 Exclusion restriction: Shock orthogonality condition

Identification in my setting relies on the exogeneity of the shocks; tariffs are ‘as-good-as-randomly assigned’ to industries from 1991 to 1997; an industry-level orthogonality condition, which I motivate from starting with the standard exclusion restriction for an instrument.

Let z_{dt} denote the shift-share instrument, and $w_{dj} = \frac{L_{dj91}}{L_{d91}}$. The exclusion restriction from the reduced-form specification (2) can be written as follows, given two time periods:

$$Cov(z_{dt} - \bar{z}_d, \epsilon_{dt} - \bar{\epsilon}_d) = Cov(z_{dt} - \frac{1}{2}(z_{dt_1} + z_{dt_2}), \epsilon_{dt} - \frac{1}{2}(\epsilon_{dt_1} + \epsilon_{dt_2})) = Cov(\Delta z_d, \Delta \epsilon_d) = 0$$

The population covariance between the SSIV and untreated potential outcomes can be shown to yield the orthogonality condition (refer to appendix 8 for the expanded derivation, adapted from Borusyak et al. (2018)). The proof and consistency of the SSIV estimate does rely on the Law of Large Numbers. With 387 cross-sectional units, at the least the weak Law of Large numbers would plausibly hold, yielding convergence in probability to the sample mean - i.e. $(\sum_{j=1}^J w_j \Delta \text{Tariff}_{j,97} \Delta \phi_j) - Cov(\Delta z_d, \Delta \epsilon_d) \xrightarrow{p} 0$.

¹⁵Whilst controlling for the share separately means this variation is swept out by the Frisch-Waugh-Lovell theorem, perhaps the treatment coefficients are also more economically meaningful compared to when renormalising the shares, given it is likely that the proportion of the total workforce that is exposed to tariffs (rather than the denominator being only workers exposed) is directly also affecting the outcomes, and hence part of the treatment of interest.

$$Cov(\Delta z_d, \Delta \epsilon_d) = \mathbb{E}(\sum_{j=1}^J w_{dj} \cdot \Delta \text{Tariff}_{j,97} \Delta \epsilon_d)$$

By the Law of Large Numbers,

$$\begin{aligned} &= \frac{1}{D} \sum_{d=1}^D \sum_{j=1}^J w_{dj} \cdot \Delta \text{Tariff}_{j,97} \Delta \epsilon_d \\ &= \sum_{j=1}^J w_j \Delta \text{Tariff}_{j,97} \Delta \phi_j \rightarrow 0 \end{aligned}$$

where $\Delta \phi_j = \frac{\mathbb{E}(w_{dj} \Delta \epsilon_d)}{\mathbb{E}(w_{dj})}$ is an exposure-weighted average of the change in unobserved factors affecting; e.g. employment growth of regions, in industry j .

Thus, shock-exogeneity can be motivated with the assumption:

$$\mathbb{E}(\Delta \text{Tariff}_{j,97} \mid \Delta \phi_j) = \mu \quad \forall j \quad (3)$$

In words, the necessary condition for exogeneity is that the change in tariff between 1991 and 1997 for each industry should be the same in expectation for all industries, conditional on the average change in ϵ_d across regions weighted by the industry-exposure share w_{dj} . This is a slightly weaker assumption than the pure randomisation of tariffs to industries (stronger assumption satisfying (3)); the shift-share regression will provide consistent estimates given tariff changes between 1991 and 1997 were not systematically higher or lower in industries concentrated in regions which experienced shocks to labour supply and factors influencing marriage and fertility.

An alternative path to exogeneity of shift-share instruments is assuming exogeneity of the shares. In my context, this would require the assumption that regions with different industry compositions would evolve similarly in employment, marriage and fertility, absent of trade liberalisation reforms. This is an implausible assumption, since industry-employment shares are generic in nature. The shares are likely not exogenous to unobserved outcome trends and would also capture a region's exposure to other labour supply or demand shocks on a region. Hence, I assume shock-exogeneity since a share-weighted average of random shocks is still itself as good as random.¹⁶ Shock-exogeneity as a sufficient condition for exogeneity of a shift-share instrument is also made more explicitly by transforming the regression from location-level to shock-level through the interchangeability of summation operators, also highlighting how each industry's shock is weighted. An adaptation of the proof to my context for greater intuition of the SSIV, showing the equivalence of the coefficient of interest between the region-level and industry-level regressions, can be found in 8.

¹⁶Bartik describes how since the shift-share IV uses shock-weighted industry shares as an instrument, as opposed to industry shares themselves, it is plausible that the SSIV is itself uncorrelated to local factors even if local industry shares are themselves not (Tim Bartik's commentary on this World Bank blog (McKenzie 2018))

Remark on what SSIV is estimating: Adão et al. (2019) and Borusyak et al. (2018) prove that when assuming exogenous shocks, SSIV regressions with expected heterogeneous treatment effects (different regions' employment responds differently to the overall tariff exposure metric) estimate a convex average of location-specific linear effects (weighted mean, where weights are non-negative and sum to 1). This assumes a monotonicity condition, which is similar to that shown by Angrist and Imbens (1994) in their seminal IV paper which allows for a Local Average Treatment Effect to be identified, but considering both industry- and location-specific effects of shocks to a region's treatment value, as opposed to a region's instrument and treatment values alone.

5.3 Gender-specific SSIV

The overall degree of trade protectionism by region can be decomposed into gender-specific exposure. A given reduction in tariffs for a particular industry would differentially affect male and female economic stature, depending on the employment composition of the industry exposed.

To construct gender-specific trade exposure by regions, I interact the start-of-period industry-employment share with the start-of-period gender share for that industry, which comes down to the number of workers L in industry j of gender f or m ¹⁷, in region d over the main number of workers in the region, similar to the approach taken by Autor et al. (2019).

The instrument industry share for female workers: $\frac{L_{dj91}}{L_{d91}} \cdot \frac{L_{dfj91}}{L_{dj91}} = \frac{L_{dfj91}}{L_{d91}}$, where treatment share: $\frac{L_{dfjt}}{L_{dt}}$

Reduced form:

$$Y_{dt} = \alpha + \beta_1 \left(\sum_j \frac{L_{dfj1991}}{L_{d1991}} \cdot \text{Tariff}_{j1991,1997} \right) + \beta_2 \left(\sum_j \frac{L_{dmj1991}}{L_{d1991}} \cdot \text{Tariff}_{j1991,1997} \right) + \gamma_d + \lambda_t + \kappa \left(\sum_{j \neq OS} \frac{L_{dfj1991}}{L_{d1991}} \right) \delta_1 + \kappa \left(\sum_{j \neq OS} \frac{L_{dmj1991}}{L_{d1991}} \right) \delta_2 + \epsilon_{dt} \quad (4)$$

I again control for a linear time trend in the sum of baseline shares, as in (2), for the proportion of each gender's workers exposed to treatment. β_1 now captures the partial effect of tariff shocks affecting women employed, holding fixed the shock to men employed, and vice versa for β_2 . By comparing β_1 and β_2 , we can also gauge relative effects - i.e. whether female-intensive exposure has a larger effect than male-intensive exposure.

¹⁷The Indian census only records data by binary genders.

5.4 Potential limitation - computation of standard errors

An inference problem with shock-level identification is that regions with similar industry employment compositions will mechanically have similar instrument values, and so may have correlated error terms through other labour demand and supply shocks to industries. This would introduce serial correlation in the error terms that clustering at a conventional geographical level does not resolve, since we assume regression residuals are independent between clusters (regions) (Adão et al. 2019). Thus, there is a possibility of underestimating standard errors, and over-rejection of the null hypothesis in my region-level estimates. Estimating the regression at the shock-level (industry) with usual robust standard errors would yield identical coefficients, while producing appropriate exposure-robust standard errors. The Stata package `ssaggregate` designed by Borusyak, Hull and Jaravel helps implement this, by transforming the dataset and variables from location-level to industry-level, where each variable is now weighted by average industry-exposure shares. To consider the weights for each industry capturing the importance of the shock¹⁸, I implement the package, but with 9 industry groups (which are facing treatment shocks - Other Services does not) this results in 18 observations in total, which yields inconsistent results and a weak first-stage.

I considered using data disaggregated into further industries which would be available at a higher regional level (at state) and then running industry-level regressions.¹⁹ I feel the industry-level regression alone would not have had much clear intuitive interpretation; given the equivalence between location-level and region-level coefficients, answering the research question would come back to considering the coefficient in the regional setting. Estimating regionally at a higher aggregated level would result in different smaller areas' exposure being cancelled out and yielding limited variation in the treatment at the state-level. Hence, my estimation continues at the region-level as opposed to the industry-level.

6 Results

6.1 Overall exposure

My shift-share instrument is strong, with an F-statistic of 617 and coefficient of 1.43, for the urban regions of interest. Figure 2 below plots the relationship between predicted shift-share treatment values from the first-stage regressions and the actual treatment values, where the marker size corresponds to the population of each region. This is plotted for 2001, since the treatment and instrument values are identical in 1991.

¹⁸In the equivalence proof between the location-level to industry-level regression in appendix 8, the industry-level regression is weighted by w_j ; the average exposure of locations to industry j

¹⁹India's Annual Survey of Industries datasets offers employment data disaggregated by further industries at the district level. At the start of my project, I was able to access the respective 1990-91 files, however regrettably ran into difficulties of removed Nesstar files when attempting to export 2000-2001. Indeed, data access was sometimes difficult throughout the project, with the census website servers occasionally refusing to connect to devices outside of India.

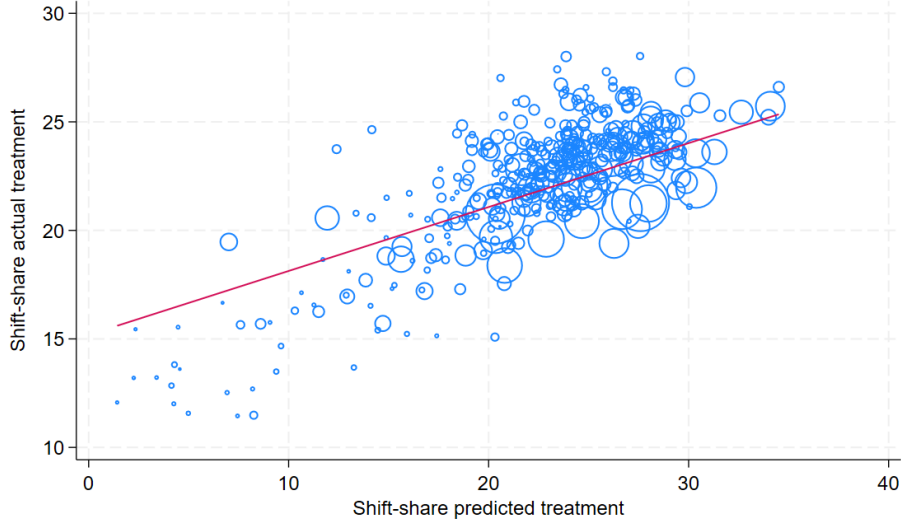


Figure 2: Overall exposure SSIV results

Table 1 presents the coefficient estimates from SSIV regression (2). It is worth restating the interpretation of the treatment coefficient; it represents the effect of an increase in a region's overall degree of trade protectionism by tariffs, controlling for a linear time trend in the share of baseline workers (out of all main workers) in traded industries - i.e. the larger the treatment value, the more workers there are in industries with higher tariffs. For ease of interpretation, I have rescaled instrument and treatment coefficients throughout to represent a 10-unit change in the shift-share treatment, which for overall region exposure roughly corresponds to one standard deviation of the treatment (9.66).

Table 1: Overall exposure SSIV results

	(1)	(2)	(3)	(4)	(5)	(6)
	Overall	Female	Male	Marriage	Marriage	Fertility
	employment	employment	employment	(15 - 19)	(20 - 24)	(20 - 24)
SSIV treatment	0.000	0.017**	-0.007	0.006	-0.030**	-0.011
(coefficients *10)	(0.008)	(0.008)	(0.010)	(0.013)	(0.014)	(0.009)
Sum of baseline shares	-0.049***	0.042**	-0.109***	-0.103***	-0.200***	-0.045**
(time trend control)	(0.018)	(0.019)	(0.023)	(0.031)	(0.034)	(0.022)
Dep. var mean	0.366	0.115	0.581	0.197	0.673	0.121
(s.d)	(0.047)	(0.059)	(0.062)	(0.010)	(0.128)	(0.052)
N	387	387	387	387	387	387

SSIV estimates, regression (2), of overall regional exposure to trade on outcomes (resized coefficients by s.f10). Regional and time fixed effects, controlling for sum of baseline shares. Clustered standard errors, at regional level, in parentheses. 774 region-time pairs.

As a region's workers are more protected from trade, female employment increases, statisti-

cally significant at the 5% level; an increase in the degree of trade protectionism by one standard deviation results in a 14% increase²⁰. A one standard deviation increase in SS treatment corresponds to a 4% decrease in the share of 20 - 24 year-old women married. A small positive but statistically insignificant effect is estimated on the share of women married from ages 15 - 19. Although large confidence intervals reflect a degree of uncertainty in the precise magnitude of effects, my results suggest urban regions which were more exposed to trade *liberalisation* reforms experienced lowered female employment rates, and likely related increases in the marriage rates for a subset of young women. This rise in marriage is consistent with Shruti Sengupta’s findings for the marriage rates for women between 15 - 34 years. Results by overall exposure thus support the marriage mechanisms discussed in 3 as part of the Becker model of household specialisation, where improvements in a women’s outside option to domestic work deters marriage, and a worsening of economic opportunities for women increases the attractiveness of marriage.

Male employment has a modest (in relation to the mean) negative coefficient, suggesting greater exposure to trade liberalisation reforms marginally increased male employment. This might lend support to the idea of there being a converse relationship between male and female labour force participation, but I fail to reject that the coefficient is statistically different to zero. I estimate greater exposure to have increased fertility for 20 - 24 year old women, moving in the same direction as marriage rates, but this also is imprecisely estimates and statistically insignificant.

The coefficients for the incomplete shares control broadly imply that an increase in the baseline share of workers *not* in ‘Other Services’ contributes to a fall in male employment, marriage and fertility, consistent with the prior discussion of regions with greater shares of workers in ‘Other Services’ likely experiencing increases in employment, potentially due to gains in human capital or regional infrastructure. This relationship going the opposite way for female employment lends suggestive credibility to the ‘added worker’ effect and converse relationship between female employment and early marriage rates.

Broadly, I would expect the overall exposure results weight male exposure more heavily, given 86% of the workforce is male on average. I consider gender-specific exposure to test for more nuanced heterogeneous and relative effects, holding constant the shock to the other gender in the region’s labour force and appropriate controls for the share of each gender in traded industries.

6.2 Gender specific exposure: employment

My first stages for the gender-specific endogenous treatment on the corresponding instrument, after partialling out the other gendered instrument, are sufficiently strong (F-statistic of 418

²⁰ $\frac{1.679}{10} \times 9.66 = 1.623$ - 1 s.d. increase in SS treatment corresponds to 1.62 pp increase in female employment.
 $\frac{1.623}{11.52} = 14.1\%$

for male-specific SSIV, and 71 for female-specific SSIV), implying there is sufficient variation between the instruments to separately identify dynamics. The instruments also have a correlation of 0.437, further alleviating high multicollinearity concerns. Female-specific shift-share treatment and instrument values are considerably lower than male-specific, given low levels of female labour force participation resulting in small female-industry shares.

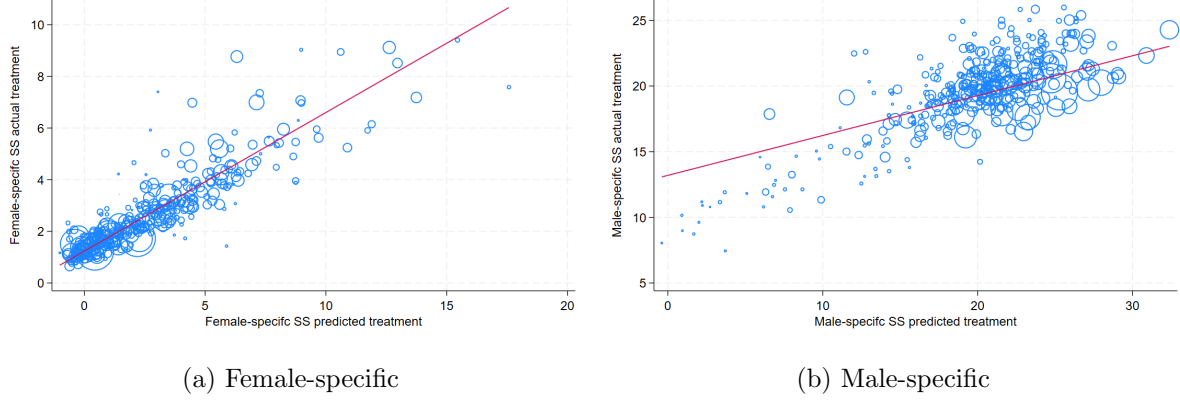


Figure 3: Predicted SS treatment from first-stage, by gender

Figure 4 presents the estimates of the effect of a region’s gender-specific exposures to trade protectionism on employment outcomes. Each panel presents the estimates from one singular regression (4). The coefficient on female specific exposure can be interpreted as the effect of increasing women’s exposure to trade protectionism (more working women in industries with higher tariffs), holding fixed male workers’ exposure in a region and controlling for separate time trends in the shares of working women and men in trade-exposed industries at baseline.

As working women’s exposure to trade protectionism in a region rises by one standard deviation, holding fixed men’s exposure, female employment decreases by 32%, statistically significant at the 5% level²¹.

Finding a positive coefficient from overall regional exposure (in regions with more exposure to *trade liberalisation*, female employment *fell*), and a negative coefficient of female-specific exposure on female employment (*positive* effect from *liberalisation*) seems counter-intuitive at first pass. However, I believe this can likely be explained by the effects on female employment being heterogeneous by different sectors of workers. As a region became more exposed to trade liberalisation reforms, through general equilibrium effects, women working in the non-traded ‘Other Services’ sector may have been negatively impacted and experienced increases in unemployment. Indeed, when I add the same linear time trend explicitly controlling for the share of women working in traded industries to the overall exposure regression, the shift-share treatment coefficient is statistically insignificant and modestly negative.

Trade liberalisation reforms benefitted women working in traded industries, hence controlling for a time trend for women working in these industries at baseline (and so holding constant the share of working women in Other Services), a region with greater female exposure to trade

²¹ $\frac{-0.121}{10} \times 3.011$ (s.d of female-specific SS treatment) = -0.0364. $\frac{-0.0364}{0.1152} = -0.317$

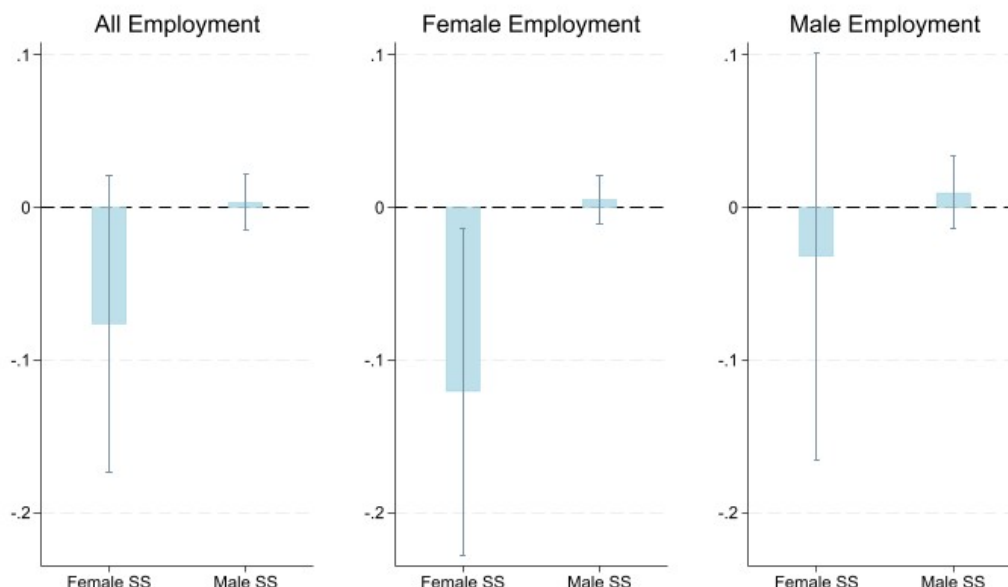


Figure 4: Gender-specific SSIV results for employment, resized coefficients by s.f10

Point estimates (clustered SEs) for: All employment: Female SS = -0.076 (0.050), Male SS = 0.003 (0.009). Female employment: Female SS = -0.121 (0.055), Male SS = 0.005 (0.008). Male employment: Female SS = -0.032 (0.068), Male SS = 0.010 (0.012). 95% confidence intervals shown.

liberalisation experienced gains in female employment. Since 42 - 45% of working women across the time period worked in ‘Other Services’, in the overall exposure effect on female employment, the impact on ‘Other Services’ is more heavily weighted and overshadows positive effects of trade liberalisation for women in traded industries.

Both the estimates of male-specific exposure and the overall regional exposure to trade reforms would indicate little, if any, effects on male employment. Estimating the effect of the male-specific shock alone without controlling for female-specific exposure does not change the significance or direction of the coefficient, confirming the small and insignificant estimates from the male-specific shock are not driven by partialling out female-specific exposure. This result makes it difficult to test the possibility of an added worker or income effect, as motivated ex-ante. The question remains of why we see significant effects on women’s employment, but not on men’s. While I do not arrive at a conclusive explanation, considering how shocks to specific industries contribute to overall results in Section 7.1 offers greater insight.

6.3 Gender-specific exposure: marriage and fertility

Figure 5 presents the gender-specific SSIV coefficients for the share of 15 - 19 and 20 - 24 year-old women married, and the share of 20 - 24 year-old women who gave birth in the previous years. An increase of one standard deviation in the female-specific degree of exposure to trade protectionism (liberalisation) corresponds to a 34% increase (decrease) in the share of 15 - 19

year-old women married.²² Interestingly, the significant effect is seen on the marriage share of 15 - 19 year-old women, as opposed to the overall exposure affecting the marriage share of 20 - 24 year-old women.

Consistent with the story above of the female-specific exposure estimate in (4) more heavily capturing the effect of women in traded industries, women working in these industries would be more likely to marry earlier compared to women in ‘Other Services’, who have a greater probability of having obtained a higher education qualification instead. Thus the gains to the sector of female workers in traded industries causes the marriage rate to fall for a younger subset of women. The negative relationship between changes to female employment and marriage rates remains consistent, supporting that a rise in economic opportunities for women drives an independence effect, which on average dominates any potential dowry effect. The nature of the trade liberalisation reforms as an exogenous labour market shock interestingly informs of the effect specifically in this direction - the relationship is also in theory reversely causal. Earlier marriage is documented to reduce the education obtained by women, and lower human capital development limits economic opportunities and freedom available. Since tariff reductions are plausibly unrelated to marriage through other outcomes, we can concretely see employment affecting marriage.

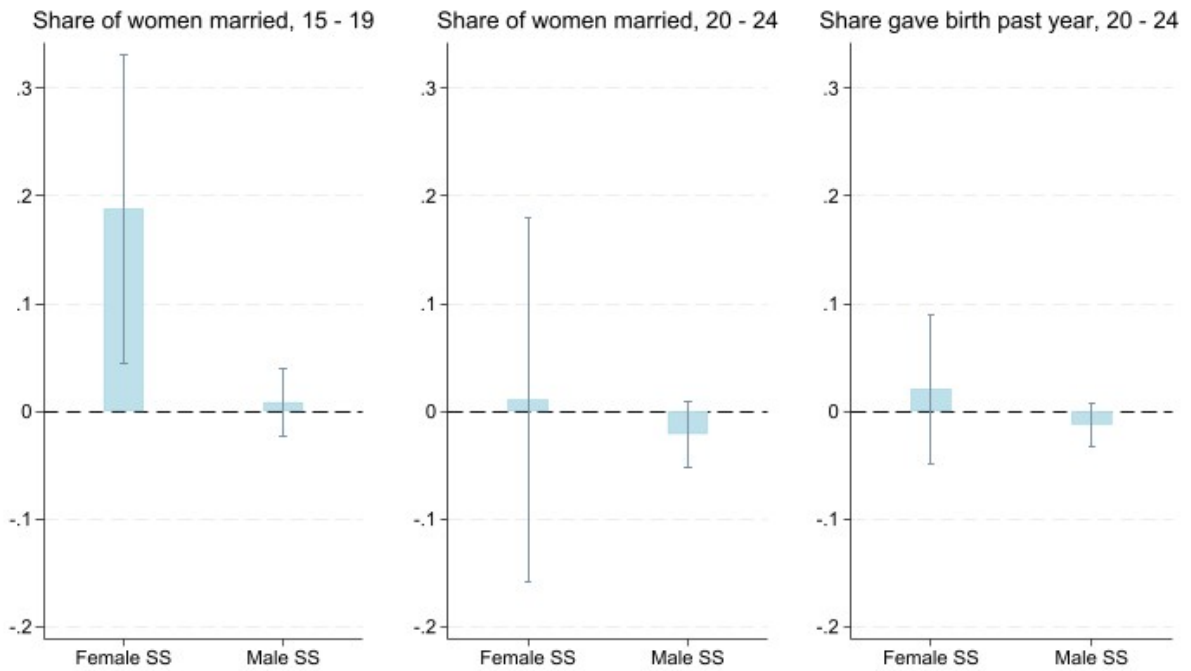


Figure 5: Gender-specific SSIV results for marriage and fertility share. 95% confidence intervals shown.

The effect of male exposure to trade liberalisation on both marriage and the share of women giving birth is positive but close to 0, in line with there being little effect on male employment. The female-specific coefficient on the share of women who gave birth is in a similar direction to

²² $\frac{0.1883}{10} \times 3.011 = 0.0567$, $\frac{0.0567}{0.197} = 0.339$

the marriage rates, but at a much smaller magnitude and the null hypothesis of there being no effect cannot be rejected. The significant effects seen on the rates of women marrying earlier, but not on the share of women giving birth may reflect the degree of resistance of social norms around both outcomes, where the frequency of married women having children is less responsive to changing economic circumstances.

Disparities between estimates for the effect of a region’s overall exposure and gender-specific exposure suggest significant heterogeneities in how workers were affected and the mechanisms behind results. My results are suggestive of the impacts of trade liberalisation varying depending on the subset of workers for who outcomes are considered, and the composition of workers exposed to trade reforms. Hence, I next consider how each industry group contributes to full estimates.

7 Potential mechanisms and robustness checks

7.1 Driven by particular industries? Leave-One-Out construction

To consider how the shock to each particular industry is driving the results, I implement leave-one-out construction of the shocks (in a similar vein to jackknife instrumental variable estimation). Each industry group is iteratively left out from the construction of the instrument and treatment and I alter the incomplete shares control to account for the omitted industry to estimate the regression with the remaining industries. In essence, running the same regression (2) but replacing j with $j - 1$ industries in nine different regressions (since ‘Other Services’ is not a component of the treatment). Figures 6 and 7 show the coefficient plot with the estimates, regressing female and male employment on the leave-one-out construction of a region’s exposure to trade protection - where the industry label refers to the industry left out (refer to Appendix for remaining results).

My main results are consistent with leaving out majority of the industry groups, for both regressing male and female employment. The two clear exceptions are manufacturing, and wholesale, retail and trade, which are intuitively the two industries we’d expect to be most affected by India’s trade liberalisation reforms. Both industries have the highest shock-importance weighting (computed with `ssaggregate` - see Appendix: manufacturing = 0.10; wholesale, retail and trade = 0.148²³), derived from having large employment shares on average.

The manufacturing industry shock appears to drive employment estimates down (i.e. leaving out manufacturing would result in a higher shift-share coefficient estimate) for both male and female employment, and Wholesale, Retail and Trade drives the coefficient estimate up. In other words, trade liberalisation shocks to the manufacturing industry caused employment to increase, and shocks to wholesale, retail and trade caused employment to decrease.

²³i.e. if an industry had a shock importance of 1, it would reflect that it is essentially the only industry contributing to treatment

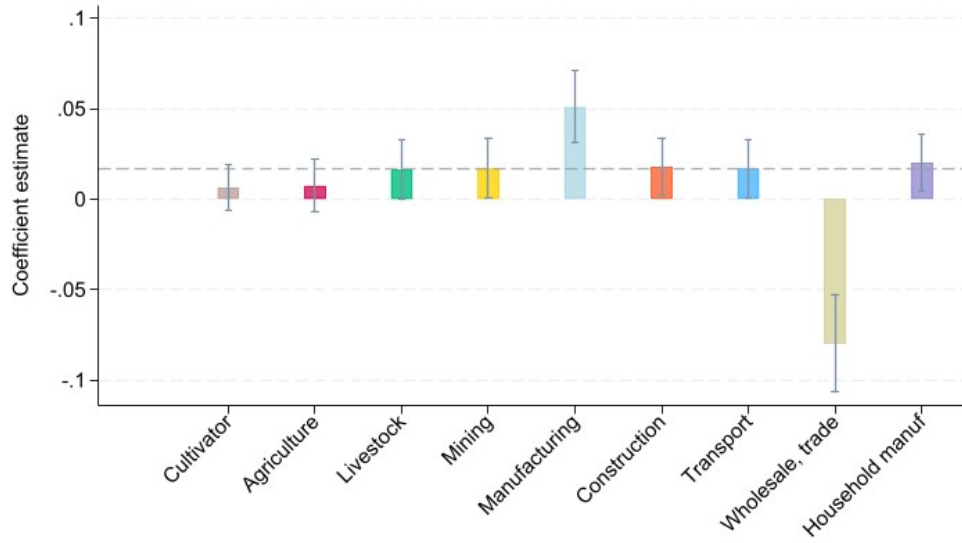


Figure 6: Estimating SSIV **female employment** regressions, iteratively leaving out industry group
Dashed grey line indicating estimate for full sample

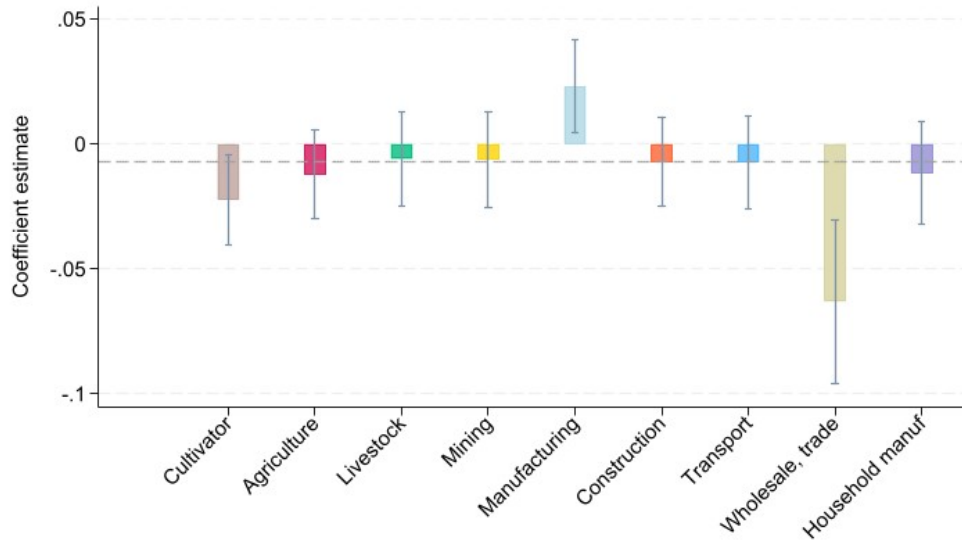


Figure 7: Estimating SSIV **male employment** regressions, iteratively leaving out industry group
Dashed grey line indicating estimate for full sample

These results are consistent with the literature for the Indian context. Topalova and Khandelwal (2011) find firm-level productivity increased from liberalisation measures through both pro-competitive effects, where lowered final-good tariffs caused the average domestic firm to become more productive, and from having access to cheaper, superior inputs which had a stronger effect. The manufacturing industry was the most heavily protected in 1991 (average AHS import

tariff of 76.96%) and experienced the largest decline in tariffs (26.02% in 1997, and 27.46% in 2001). Wholesale, retail and trade had the lowest tariffs among the industry groups in 1991 (24.49%), meaning the industry was relatively already significantly exposed to trade. While in line with the rest of the industries where tariffs fell from 1991 to 1997 (13.15%) and then rose from 1997 to 2001; the average tariff for wholesale, retail and trade rises from 1997 to 2001 to the extent that the tariff is higher in 2001 (38.6%) than it is in 1991. Given Wholesale comprises of selling and arranging the purchase of goods for resale, I would expect the industry to benefit less from cheaper inputs to enhance production processes. While they could resell cheaper foreign imports, if primarily handling domestic goods, the industry would face even greater competition.

The effects driven by particular industries can also point towards the limited average effect on male employment. Manufacturing and wholesale, retail and trade can account for greater shares of working men (14% and 25% respectively in 2001) than for women (approximately 9% for both). Relatedly, looking at the results for the gender-specific regressions (Fig 11 and 12 in the Appendix), the coefficient on male-specific exposure leaving out these industries is significantly higher (and statistically significant) relative to female-specific exposure, actually informing of greater effects of men’s exposure to these industries in contradicting directions.

A potential concern from the fact that particular industries individually have a significant effect on overall estimates is that the exogeneity of the shift-share instrument might be compromised. A baseline assumption underlying exogeneity via the shocks is that the number of shocks is large, and so in this case, that there are sufficiently many industries. Census data has employment at the district level by ten industry groups, which is not ideal. The concern with having fewer shifts is that they might by chance be correlated with unobservable characteristics (ϕ_j above) even if ‘randomly assigned’. I argue that while what we observe with my dataset is shocks to these nine aggregated industry groups, these have been averaged over many shocks (changes to tariffs for over 5000 product lines). In a similar logic to the shift-share instrument itself, averaging over many ‘independent’ shifts to form an aggregated measure would mean the single value for the industry is uncorrelated with unobservable share-weighted regional characteristics.

7.2 Migration

I also note that migration could potentially dampen regional labour market effects; for example, if people struggling to find employment move from more heavily affected regions to those less affected, then the regional effects of changes in tariffs would be underestimated (in absolute terms), since the employment rates documented in a region would not reflect those unemployed who move elsewhere - i.e. a region where workers lost jobs as a result to exposure to trade liberalisation reforms, but many of those workers over the course of the decade relocated to other districts or states, would not report much change in the employment rate, masking this effect.

Given the shift in economic policies in 1991 might have had a significant effect on migration from the low baseline rates, I estimate the SSIV regressions controlling for migration, adding as controls both: 1) the share of people who migrated to an urban part of a different district within the state between 1991 and 2001 (out of total population in 2001), interacted with a 2001 period indicator, and 2) the share of people who migrated to an urban region from another state in India interacted with the 2001 indicator. My results are shown in the appendix and are robust to controlling for inter-state and intra-state migration, but the effects of male-specific exposure on marriage and fertility outcomes for 20 - 24 year-old women become statistically significant at the 10% level, while remaining at similar point estimates.

7.3 Large regions

My construction of composite regions, where boundaries are unchanged from 1991 to 2001, exacerbates significant existing variation in population sizes across regions. With large numbers of workers and diversified smaller areas, the shift-share treatment and instrument values would be closer to the national average values. I check to see how dropping these regions with populations past the 90th percentile affects my results (dropping 32 regions from my sample). My results are broadly robust to this, with the exception of the marriage rate for 20 - 24 year old women which is no longer statistically significant, but a similar sized negative coefficient is estimated (-0.023 compared to the full sample estimate of -0.030, mean of 0.676). Male-specific exposure estimates are modestly positive and become statistically significant for outcomes: male employment and the marriage rates for 15 - 19 year-old women. A one unit increase in the degree of male-specific exposure to trade liberalisation led to a 4.2% decrease in male employment and 18.7% decrease in marriage rates for 15 - 19 year-olds, supporting Becker's theorised impacts on marriage becoming less desirable with a worsening of men's economic opportunities in this partial sample.

7.4 Balance check: past related factors

Ideally, to check the balance of the shift-share instrument at the local regional level, I would have liked to regress past employment, marriage and fertility rates on the SSIV. This would allow me to consider whether changes in tariffs were systematically different in industries concentrated in regions which had unobserved confounders affecting outcomes of interest prior to the treatment. While the Indian census site does not have data available from censuses prior to 1991, I utilise district-level data from the International Crops Research Institute for the Semi-Arid Tropics (ICRISAT) to regress the change in a plausibly related factor - the number of banks per district, between 1991 - 2001, and 1976 - 1986²⁴, on the change in the region's exposure to treatment - estimating regression (2) as a first-difference.

I find a statistically significant effect on the shift-share treatment coefficient for the change in region's number of banks from 1991 to 2001 at the 5% level, but not for the change in number

²⁴The years 1976-86 were selected, since they had the most number of observations prior to 1991

of banks from 1976 - 1986. Unfortunately, this is estimated with 56 regions (112 observations)²⁵, which raises concerns of consistency and the standard errors being biased downwards. I consider these suggestive results to cautiously support the exclusion restriction made.

8 Conclusion

Through a shift-share instrumental variable strategy, I leverage regions' differing pre-determined exposure to national changes in tariffs to consider how the degree of exposure to trade liberalisation reforms affected long-run regional employment, marriage and fertility rates. Amidst the current backdrop of increased trade uncertainties and shifting relations for both developed and developing countries, assessing impacts and potential dynamics at a regional level proves pertinent, to understand consequences on geographical inequalities and guide local policies and interventions. I find that an increase of one standard deviation of a region's degree of exposure to trade liberalisation reforms in Urban India corresponded to a 14% decrease in female employment, and a 4% increase in the marriage rate for women aged 20 - 24 years old. While I find a small increase in the men's employment rate, I fail to reject that this differs from no effect.

Considering the effects of gender-specific shocks suggests heterogeneous impacts from the reforms for subsets of workers, with economic opportunities for women working in non-traded services potentially having worsened while those working in traded-industries gained. Analysis iteratively dropping industries confirms this effect, with tariff reductions to industrial manufacturing having positive effects on employment, and the impact on wholesale, retail and trade having a negative effect. Varying long-run impacts by sectors contributes to explanations of widening disparities for workers and regions.

I consistently find a similar mechanism of women's employment opportunities moving negatively with marriage rates at ages 15 - 19, implying the 'independence effect' for women from greater economic opportunities on average outweighs any potential 'male backlash' or dowry effect. The degree of gains from female employment in the trade liberalisation context also appears to have outweighed the social cost around the stigma of working women. This has important implications supporting efforts to broaden access to formal employment for women, given delayed marriages in developing countries has a substantial impact on their welfare, as well as wider positive externalities to outcomes for society.

²⁵This is largely due to the ICRISAT having datasets either consistent with 1966 district boundaries onwards, or 2011 onwards. Since my main dataset has regions consistent with 1991 district boundaries, I keep regions which had the same boundaries in 1966 and 1991.

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A1: Proofs

A1.1 Location-level and industry-level coefficient equivalence

Proof. The coefficient estimates from the location-level and industry-level regression are numerically equivalent

Starting with the location-level reduced form analysis: L_{dj} is the number of workers in region d in industry j :

$$Y_{dt} = \alpha + \beta \left(\sum_j \frac{L_{dj1991}}{L_{d1991}} \cdot \text{Tariff}_{j1991,1997} \right) + \gamma_d + \lambda_t + \kappa \left(\sum_{j \neq OS} \frac{L_{dj1991}}{L_{d1991}} \right) \delta + \epsilon_{dt} \quad (5)$$

- Let $\frac{L_{dj1991}}{L_{d1991}}$ be written as w_{dj} : pre-determined industry-employment share.
- Let $x = \left(\sum_j \frac{L_{dj1991}}{L_{d1991}} \cdot \text{Tariff}_{j1991,1997} \right)$ - endogenous treatment variable being instrumented by $w_{dj} \cdot \text{Tariff}_{j1991,1997}$
- Let $v_{dt}^\perp$ denote the residualised variable (on fixed effects and incomplete shares control) v .

By the Frisch-Waugh-Lovell theorem, the location-level SSIV estimator is:

$$\hat{\beta} = \frac{\frac{1}{D} \sum_{d=1}^D \sum_{t=1,2} \left(\sum_{j=1}^J w_{dj} \cdot \text{Tariff}_{j91,97} \right) Y_{dt}^\perp}{\frac{1}{D} \sum_{d=1}^D \sum_{t=1,2} \left(\sum_{j=1}^J w_{dj} \cdot \text{Tariff}_{j91,97} \right) x_{dt}^\perp}$$

Changing the order of the summation operators:

$$\hat{\beta} = \frac{\sum_{t=1,2} \sum_{j=1}^J \frac{1}{D} \sum_{d=1}^D w_{dj} \cdot \text{Tariff}_{j91,97} Y_{dt}^\perp}{\sum_{t=1,2} \sum_{j=1}^J \frac{1}{D} \sum_{d=1}^D w_{dj} \cdot \text{Tariff}_{j91,97} x_{dt}^\perp} \quad (6)$$

Let $\hat{w}_j = \frac{1}{D} \sum_{d=1}^D w_{dj}$, the average exposure of locations to industry j and $\bar{v}_j = \frac{\sum_{d=1}^D w_{dj} v_d}{\sum_{d=1}^D w_{dj}}$, a weighted average of v_d , by industry shares. We want to rewrite our estimate in terms of these industry-weighted average variables. To get there, we can substitute in $\sum_{j=1}^J \hat{w}_j \bar{Y}_{jt}^\perp$ for

$\frac{1}{D} \sum_{d=1}^D \sum_{j=1}^J w_{dj} Y_{dt}^\perp$, since:

$$\begin{aligned} \sum_{j=1}^J \hat{w}_j \bar{Y}_{jt}^\perp &= \sum_{j=1}^J \left(\frac{1}{D} \sum_{d=1}^D w_{dj} \right) \frac{\sum_{d=1}^D w_{dj} Y_{dt}^\perp}{\sum_{d=1}^D w_{dj}} \\ &= \frac{1}{D} \sum_{d=1}^D \sum_{j=1}^J w_{dj} Y_{dt}^\perp \end{aligned}$$

(by taking the constant $\frac{1}{D}$ out of the sum operator and cancelling out $\sum_{d=1}^D w_{dj}$)

We similarly substitute in $\sum_{j=1}^J \hat{w}_j \bar{x}_{jt}^\perp$:

$$\text{Hence, } \hat{\beta} = \frac{\sum_{t=1,2} \sum_{j=1}^J \hat{w}_j \cdot \text{Tariff}_{j91,97} \bar{Y}_{jt}^\perp}{\sum_{t=1,2} \sum_{j=1}^J \hat{w}_j \cdot \text{Tariff}_{j91,97} \bar{x}_{jt}^\perp} \quad (7)$$

Hence $\hat{\beta}$ is now equivalently the estimator from the following regression, *weighted by $\hat{w}_j$* :

$$\bar{Y}_{jt}^\perp = \alpha + \beta \bar{x}_{jt}^\perp + \bar{\epsilon}_{jt} \quad (8)$$

where $\text{Tariff}_{j91,97}$ instruments for $\bar{x}_{jt}^\perp$

□

To reiterate, $\hat{w}_j$ is the average exposure of locations to industry j , $\bar{Y}_{jt}^\perp$ is the average residualised outcome for industry j across regions

We can now more explicitly see how SSIV can be consistently estimated with exogeneity of the shocks by the ratio of the two covariances, with the necessary exclusion restriction.

Note: this proof is considering the baseline SSIV, but can be adapted to settings with multiple instruments (gender-specific interactions).

A1.2 Shock orthogonality condition

Proof. We want to derive the condition under which the ratio of covariances in (6) provides a causal effect, with the proof following a similar logic to proving industry-level equivalence.

We begin from the location-level regression. Let $\frac{L_{dj1991}}{L_{d1991}} = w_{dj}$ as in the above proof, and let z_{dt} denote the instrument for region d .

The exclusion restriction for the instrument, having partialled out fixed effects is:

$$\begin{aligned} \text{Cov}(z_{dt} - \bar{z}_d, \epsilon_{dt} - \bar{\epsilon}_d) &= \text{Cov}(\Delta z_d, \Delta \epsilon_d) = 0 \\ &= \mathbb{E}\left(\sum_{j=1}^J w_{dj} \cdot \Delta \text{Tariff}_{j,97} \Delta \epsilon_d\right) \end{aligned}$$

(w_{dj} , as a pre-determined share, remains constant)

By the Law of Large Numbers,

$$\begin{aligned}
&= \frac{1}{D} \sum_{d=1}^D \sum_{j=1}^J w_{dj} \cdot \Delta \text{Tariff}_{j,97} \Delta \epsilon_d \\
&= \sum_{j=1}^J \left(\frac{1}{D} \sum_{d=1}^D w_{dj} \Delta \epsilon_d \right) \cdot \Delta \text{Tariff}_{j,97}
\end{aligned}$$

w_j measures the expected exposure to industry j (similar to the earlier logic in A1.1, but now applying to the population), hence $w_j = \mathbb{E}(w_{dj}) = \frac{1}{D} \sum_{d=1}^D w_{dj}$. We can now multiply through by

$$\frac{w_j}{\frac{1}{D} \sum_{d=1}^D w_{dj}}:$$

$$\begin{aligned}
&= \sum_{j=1}^J w_j \Delta \text{Tariff}_{j,97} \frac{\frac{1}{D} \sum_{d=1}^D w_{dj} \Delta \epsilon_d}{\frac{1}{D} \sum_{d=1}^D w_{dj}} \\
\text{Let } \Delta \phi_j &= \frac{\frac{1}{D} \sum_{d=1}^D w_{dj} \Delta \epsilon_d}{\frac{1}{D} \sum_{d=1}^D w_{dj}} = \frac{\mathbb{E}(w_{dj} \Delta \epsilon_d)}{\mathbb{E}(w_{dj})}
\end{aligned}$$

$\Delta \phi_j$ is the exposure-weighted expectation of the change in untreated potential outcomes

$$= \sum_{j=1}^J w_j \Delta \text{Tariff}_{j,97} \Delta \phi_j \rightarrow 0 \quad (9)$$

□

Hence, shock-exogeneity can be motivated with the condition: $\mathbb{E}(\Delta \text{Tariff}_{j,97} \mid \Delta \phi_j) = \mu$

A2: Summary Statistics and Results

0.1 Summary Statistics

Table 2: Summary statistics

	1991	2001	Total	Within-region
Total population (discounted)	425,027 [752,605]	588,221 [1,043,138]	506,624 [912,618]	(179,887)
Female population (discounted)	198,200 [345,744]	271,501 [467,929]	234,850 [412,764]	(77,766)
Male population (discounted)	226,828 [407,327]	316,721 [575,736]	271,774 [500,395]	(102,366)
Overall employment share	0.385 [0.038]	0.347 [0.048]	0.366 [0.047]	(0.026)
Female employment share	0.114 [0.060]	0.117 [0.057]	0.115 [0.059]	(0.017)
Male employment share	0.616 [0.044]	0.546 [0.057]	0.581 [0.062]	(0.042)
Share of women aged 15–19 married	0.239 [0.114]	0.156 [0.059]	0.197 [0.100]	(0.055)
Share of women aged 20–24, gave birth in past year	0.129 [0.067]	0.113 [0.031]	0.121 [0.052]	(0.031)
Share of women aged 15–19, gave birth in past year	0.028 [0.028]	0.017 [0.009]	0.023 [0.022]	(0.014)
Shift share treatment	38.4 [9.66]	22.1 [3.12]	30.3 [9.66]	(8.50)
Shift share instrument	38.4 [13.5]	13.3 [2.25]	25.9 [13.5]	(12.78)
Male shift share treatment	33.6 [8.55]	19.5 [3.20]	26.5 [8.55]	(7.35)
Male shift share instrument	33.6 [11.84]	11.7 [2.14]	22.6 [11.84]	(11.12)
Female shift share treatment	4.88 [3.01]	2.66 [1.60]	3.77 [3.01]	(1.61)
Female shift share instrument	4.88 [3.15]	1.59 [1.17]	3.24 [3.15]	(2.05)

Mean summary statistics for urban observations of 387 regions. Standard deviation reported in square brackets, within-region standard deviation in rounded brackets. Population numbers exclude retirees and dependents, hence discounted.

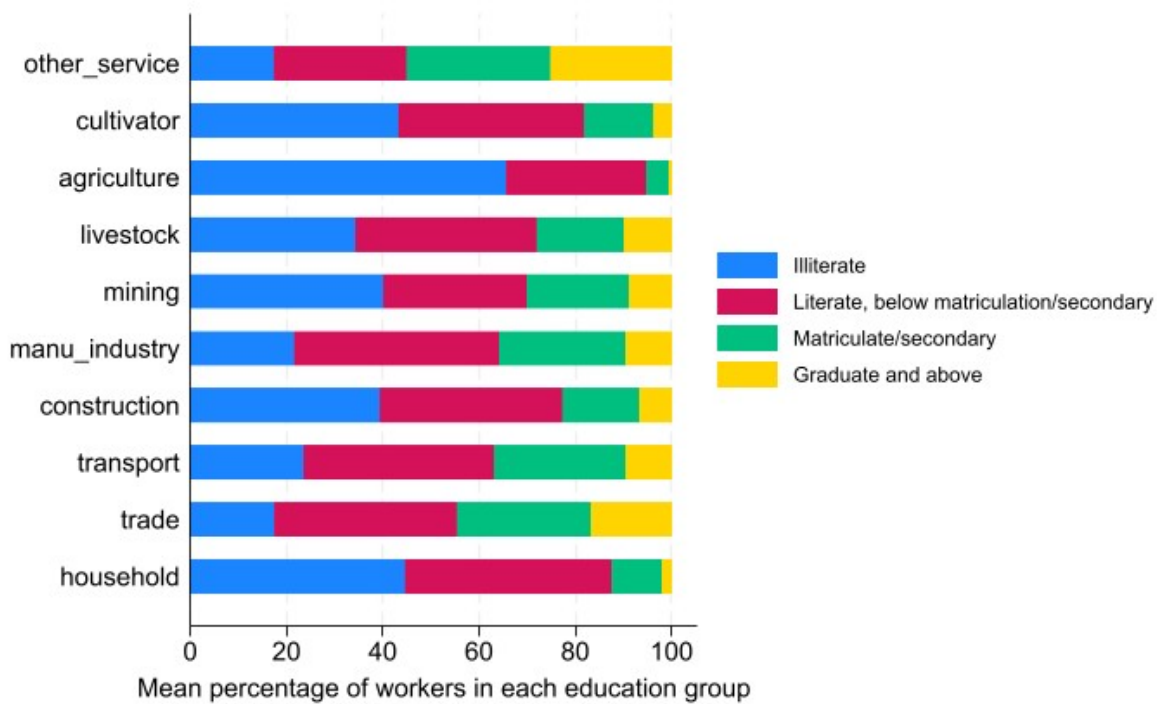


Figure 8: Mean share of workers across regions by educational qualification in each industry

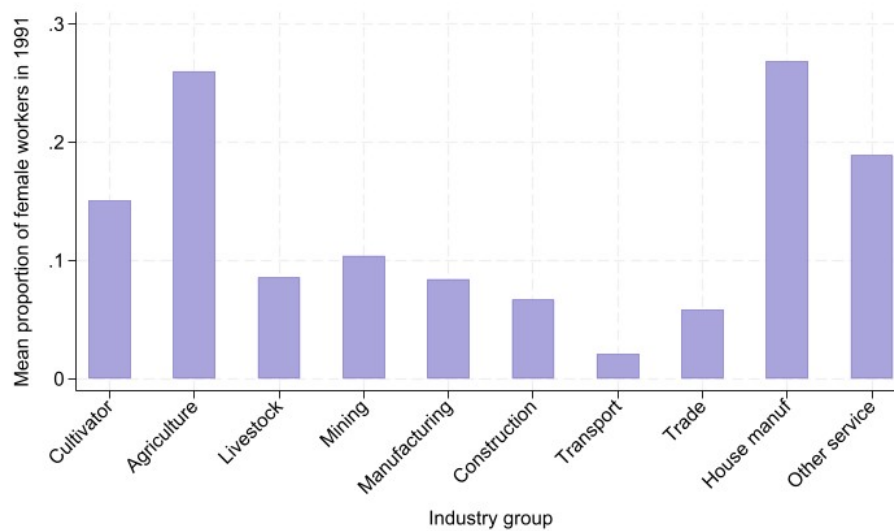


Figure 9: Gender split within each industry in 1991 - proportion of female workers

	Agricultural	Cultivators	Livestock	Mining	Manuf.	Trade	Construction	Transport	Household manuf.	OS
1991										
Female	0.174 (0.151)	0.095 (0.110)	0.018 (0.042)	0.008 (0.027)	0.090 (0.085)	0.079 (0.044)	0.026 (0.022)	0.012 (0.011)	0.073 (0.091)	0.424 (0.174)
Male	0.071 (0.054)	0.083 (0.069)	0.026 (0.030)	0.013 (0.043)	0.150 (0.089)	0.232 (0.063)	0.053 (0.028)	0.080 (0.034)	0.028 (0.034)	0.263 (0.100)
2001										
Female	0.083 (0.094)	0.060 (0.072)	0.032 (0.044)	0.013 (0.021)	0.091 (0.064)	0.093 (0.042)	0.048 (0.037)	0.017 (0.013)	0.104 (0.109)	0.459 (0.153)
Male	0.036 (0.037)	0.042 (0.040)	0.026 (0.027)	0.029 (0.041)	0.139 (0.075)	0.248 (0.059)	0.079 (0.030)	0.092 (0.030)	0.036 (0.037)	0.273 (0.094)

Mean share of total main male/female workers in each industry - i.e. each row sums to 1. Standard deviation reported in parentheses. 382 regions

Table 3: Shock importance for each industry group

Industry	Shock importance (w_j)	Tariff 1991 (%)	Tariff 1997 (%)	Tariff 2001 (%)
Cultivator	0.059	58.9	17.87	37.535
Agriculture	0.059	58.9	17.87	37.535
Livestock	0.017	47.77	8.395	24.695
Mining	0.009	50.206	17.272	20.958
Manufacturing	0.100	76.962	26.018	27.464
Construction	0.034	67.03	19.383	27.43
Transport	0.049	57.24	19.75	23.55
Wholesale, retail, trade	0.148	25.49	13.15	38.6
Household manufacturing	0.024	76.962	26.018	27.464

0.2 Robustness checks

Overall exposure: **not discounting** total population by retirees and dependents, and marriage and fertility results for all ages:

Outcome	SSIV overall	Sum baseline share
Employment	0.000 (0.006)	-0.014 (0.013)
Female employment	0.012** (0.006)	0.037*** (0.014)
Male employment	-0.005 (0.007)	-0.044*** (0.016)
Marriage (15 - 19)	0.006 (0.013)	-0.103*** (0.031)
Marriage (20 - 24)	-0.030** (0.014)	-0.200*** (0.034)
Marriage (all)	0.000 (0.004)	0.014 (0.010)
Fertility (15 - 19)	0.003 (0.004)	-0.007 (0.010)
Fertility (20 - 24)	-0.011 (0.009)	-0.045** (0.022)
Fertility (all)	-0.004 (0.005)	-0.016 (0.012)

SSIV estimates, regression (2), of overall regional exposure on employment (not discounted by retirees or dependents - simply main worker population/total population), marriage and fertility by different age groups. Resized coefficients by s.f10). 387 observations

Gender specific exposure: **not discounting** total population by retirees and dependent, and marriage and fertility results for all ages:

Outcome	Coefficient estimates			
	Female SSIV	Male SSIV	Female baseline share	Male baseline share
Employment	-0.066** (0.032)	0.003 (0.007)	-0.282** (0.120)	0.004 (0.016)
Female employment	-0.079** (0.039)	0.006 (0.006)	-0.426*** (0.140)	0.051*** (0.016)
Male employment	-0.052 (0.042)	0.007 (0.009)	-0.141 (0.164)	-0.021 (0.020)
Marriage (15 - 19)	0.188** (0.073)	0.008 (0.016)	0.736*** (0.258)	-0.140*** (0.038)
Marriage (20 - 24)	0.011 (0.086)	-0.021 (0.015)	0.054 (0.321)	-0.201*** (0.037)
Marriage (all)	0.013 (0.028)	0.005 (0.005)	0.116 (0.104)	0.017 (0.013)
Fertility (15 - 19)	0.024 (0.018)	0.003 (0.004)	0.079 (0.067)	-0.013 (0.011)
Fertility (20 - 24)	0.021 (0.035)	-0.012 (0.010)	0.085 (0.131)	-0.053* (0.027)
Fertility (all)	-0.007 (0.018)	-0.003 (0.005)	-0.021 (0.067)	-0.015 (0.013)

SSIV estimates, regression (5), of gender-specific regional exposure on employment (not discounted by retirees or dependents - simply main worker population/total population), marriage and fertility by different age groups. Resized coefficients by s.f10).

0.3 Leave-One-Out Industry results

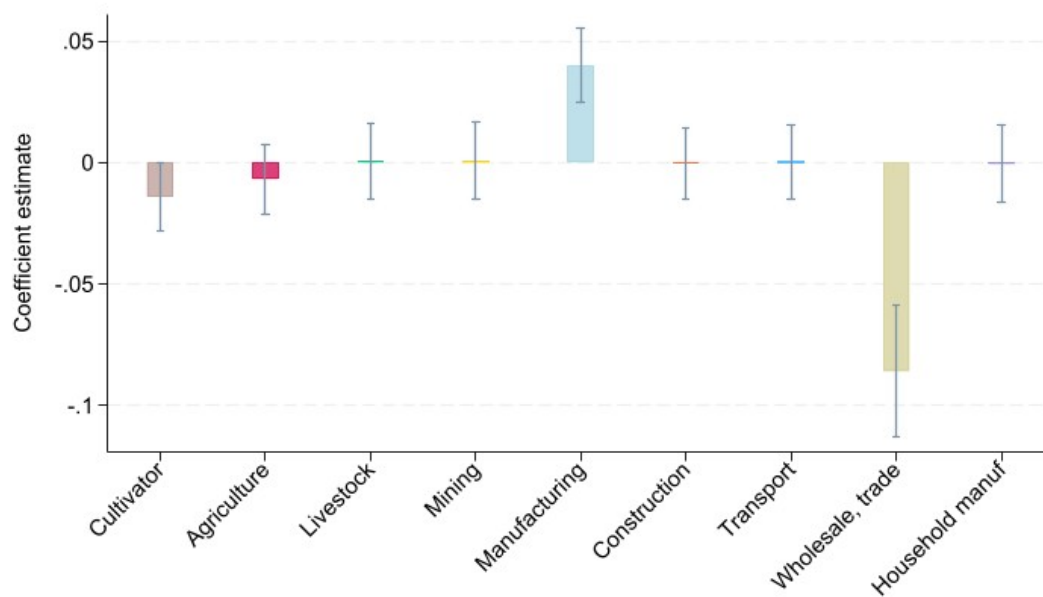


Figure 10: Estimating SSIV regressions on overall employment, iteratively leaving out industry group

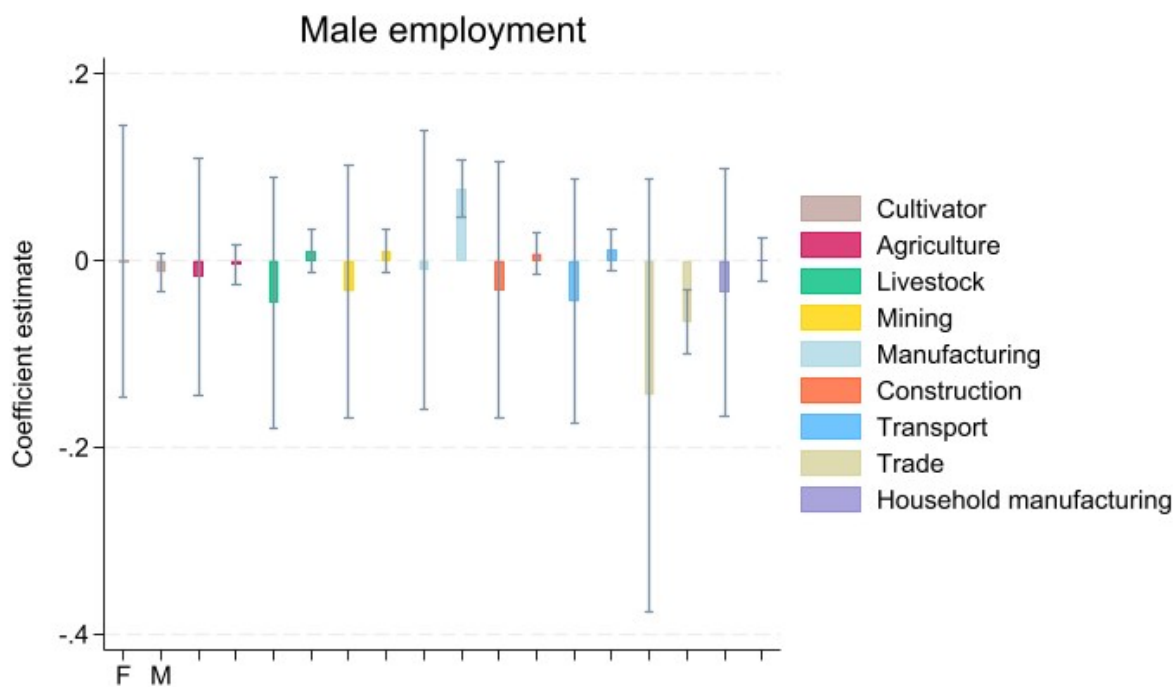


Figure 11: Estimating SSIV male employment gender-specific regressions, iteratively leaving out industry group

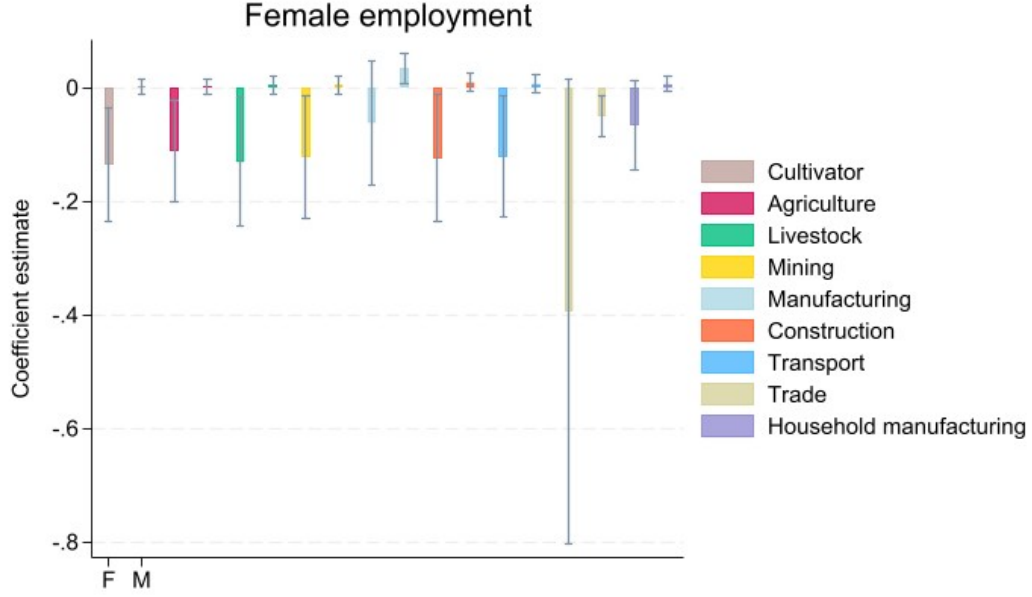


Figure 12: Estimating SSIV female employment gender-specific regressions, iteratively leaving out industry group

0.4 Robustness checks: migration controls, omitting very large regions, and number of banks

	(1) Overall employment	(2) Female employment	(3) Male employment	(4) Marriage (15 - 19)	(5) Marriage (20 - 24)	(6) Fertility (20 - 24)
SSIV treatment	-0.001 (0.008)	0.017** (0.008)	-0.008 (0.010)	0.005 (0.013)	-0.032** (0.014)	-0.012 (0.009)
Sum of baseline shares	-0.019 (0.021)	0.047** (0.019)	-0.069*** (0.025)	-0.086*** (0.031)	-0.137*** (0.035)	0.003 (0.029)
interstate_migr_control	-0.021*** (0.004)	-0.003 (0.004)	-0.029*** (0.005)	-0.017*** (0.006)	-0.042*** (0.006)	-0.026** (0.011)
intrastate_migr_control	0.002 (0.003)	-0.000 (0.002)	0.004 (0.004)	0.009* (0.005)	0.001 (0.005)	-0.008** (0.003)
<i>N</i>	774	774	774	774	774	774

SSIV estimates of shift-share treatment and controls on inter-state and intra-state migration at the state-level. Regional and time fixed effects; standard errors clustered at the regional level in parentheses. 774 region-time observations.

Table 4: Migration controls: Overall exposure results

	(1)	(2)	(3)	(4)	(5)	(6)
	Overall	Female	Male	Marriage	Marriage	Fertility
	employment	employment	employment	(15 - 19)	(20 - 24)	(20 - 24)
fshift_share_treat10	-0.075 (0.052)	-0.121** (0.055)	-0.034 (0.072)	0.167** (0.072)	0.034 (0.083)	0.069 (0.046)
mshift_share_treat10	0.001 (0.009)	0.005 (0.008)	0.007 (0.011)	0.009 (0.016)	-0.028* (0.015)	-0.020* (0.010)
interstate_migr_control	-0.020*** (0.004)	-0.001 (0.004)	-0.029*** (0.005)	-0.019*** (0.005)	-0.043*** (0.006)	-0.027** (0.011)
intrastate_migr_control	0.003 (0.003)	0.000 (0.003)	0.004 (0.004)	0.008* (0.005)	0.001 (0.005)	-0.008** (0.003)
<i>N</i>	774	774	774	774	774	774

SSIV estimates of female- and male-specific shift-share treatments with inter- and intra-state migration controls. Regional and time fixed effects; standard errors clustered at the regional level in parentheses. 774 region–time observations.

Table 5: Migration controls: gendered exposure results

	(1)	(2)	(3)	(4)	(5)	(6)
	Overall	Female	Male	Marriage	Marriage	Fertility
	employment	employment	employment	(15 - 19)	(20 - 24)	(20 - 24)
shift_share_treat10	0.009 (0.009)	0.025** (0.010)	0.000 (0.011)	0.020 (0.015)	-0.023 (0.017)	-0.008 (0.010)
sum_share_period	-0.033 (0.020)	0.057*** (0.021)	-0.095*** (0.025)	-0.078** (0.035)	-0.187*** (0.038)	-0.041* (0.024)
<i>N</i>	696	696	696	696	696	696

SSIV estimates of shift-share treatment and controls. Regional and time fixed effects; standard errors clustered at the regional level in parentheses. 696 region–time observations (dropped regions with population sizes past the 90th percentile).

Table 6: Omitting regions with populations past the 90th percentile: overall exposure

	(1)	(2)	(3)	(4)	(5)	(6)
	Overall	Female	Male	Marriage	Marriage	Fertility
	employment	employment	employment	(15 - 19)	(20 - 24)	(20 - 24)
fshift_share_treat10	-0.097*	-0.122**	-0.064	0.138**	-0.020	0.006
	(0.052)	(0.057)	(0.065)	(0.070)	(0.082)	(0.034)
mshift_share_treat10	0.019	0.012	0.029**	0.045**	-0.005	-0.006
	(0.012)	(0.010)	(0.015)	(0.022)	(0.020)	(0.013)
sum_fshare_period	-0.432**	-0.684***	-0.180	0.615**	-0.049	0.045
	(0.189)	(0.202)	(0.248)	(0.243)	(0.307)	(0.128)
sum_mshare_period	0.002	0.072***	-0.046	-0.074	-0.166***	-0.041
	(0.026)	(0.025)	(0.033)	(0.049)	(0.044)	(0.032)
<i>N</i>	696	696	696	696	696	696

Table 7: Omitting regions with populations past the 90th percentile: gender-specific exposure

	(1)	(2)
	No. banks 1991 - 2001	No. banks 1976 - 1986
D.shift_share_treat10	-23.598**	14.604
	(10.531)	(20.356)
D.sum_share_period	-79.179	284.939***
	(49.630)	(65.396)
D.total_pop	0.000***	0.000***
	(0.000)	(0.000)
_cons	27.849	-107.207***
	(31.599)	(31.259)
<i>N</i>	56	56

First-differences specification, controlling for total population in 1991 and 2001

Table 8: Testing for pre-trends in the number of banks, comparing 1991 - 2001 and 1976 - 1986